

High level policy $\rightarrow$ intent
 Low level policy $\rightarrow$ word selection.

High level MDP

- State s_t
- Action $\phi_t \in \mathbb{R}^d$
- Policy $\pi_{\text{high}}(\phi_t | s_t)$
- Transition $s_t \rightarrow (s_t, \phi_t)$
- Reward $R_{\text{high}}(s_t, \phi_t) = \text{mutual info } I(\phi_t; s_{t+1:t+k})$

Low level MDP

- State $(s_t, \phi_t) \rightarrow$ original state + intention
- Action $a_t \in V \parallel V = \text{Vocabulary}$
- Policy $\pi_{\text{low}}(a_t | s_t, \phi_t)$
- Transition Deterministic token append
- Reward $R_{\text{low}} = \text{alignment with intention} + \text{immediate fluency}$

Combined policy

$$\Rightarrow \pi(a_t | s_t) = \int \pi_{\text{high}}(\phi_t | s_t) \cdot \pi_{\text{low}}(a_t | s_t, \phi_t) d\phi_t$$